

CoT*	12.8	35.6	3.80	16.2	22.6	6.60	24.0	17.4
RAG*	27.4	58.2	17.8	25.8	23.2	9.40	16.8	25.5
Search-o1*	19.4	40.6	11.4	17.0	27.0	8.60	30.4	22.1
R1-Instruct	21.0	44.9	17.1	20.8	27.5	6.00	19.2	22.4
Search-R1	39.3	61.0	39.7	37.0	40.1	14.6	36.8	38.5
ZeroSearch	43.6	61.8	51.5	34.6	35.2	18.4	27.8	39.1
StepSearch	-	-	-	38.6	36.6	22.6	40.0	-
EvolveR	43.5	63.4	44.6	38.2	42.0	15.6	54.4	43.1
SKILLRL	45.9	63.3	45.9	43.2	40.3	20.2	73.8	47.1

Table 3. Ablation study results. We report average success rate (%) on ALFWorld and WebShop.

Method	ALFWorld	WebShop
SKILLRL	89.9	72.7
<i>Skill Library Ablations</i>		
w/o Hierarchical Structure	76.8	61.4
w/o Skill Library (Raw Trajectories)	61.7	50.2
<i>Training Pipeline Ablations</i>		
w/o Cold-Start SFT	65.2	46.5
w/o Dynamic Evolution	84.4	70.3

